

§3-1 Derivatives of Polynomials and Exponential Functions

Homework : 2 (b),(c),13,22,23,28,39,46,52,54,55,56,57,60,61,63

多項式和指數的微分：

$$(I) \quad \frac{d}{dx}(x^n) = (x^n)' = nx^{n-1}, \quad n \in N.$$

$$\textbf{Proof :} \quad f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(n x^{n-1} + C_2^n x^{n-2} h + \dots + C_{n-1}^n x h^{n-2} + h^{n-1})}{h}$$

$$= n x^{n-1}.$$

$$\textbf{Remark :} \quad \frac{d}{dx}(x^r) = r x^{r-1}, \quad r \in R, \text{ 需等到學對數的微分方可證明.}$$

$$(II) \quad (af + g)' = af' + g', \quad a \in R.$$

(微分的動作是線性(Linear)).

$$(III) \quad f(x) = a^x$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} = \lim_{h \rightarrow 0} \frac{a^x(a^h - 1)}{h}$$

$$= a^x \lim_{h \rightarrow 0} \frac{a^h - 1}{h} = a^x f'(0). \quad (3.1-1)$$

When $a = 2$, $f'(0) \approx 0.69$.

When $a = 3$, $f'(0) \approx 1.10$.

$\Rightarrow \exists$ a number, say e , $2 < e < 3$ such that

if $f(x) = e^x$, then $f'(0) = 1$

(IV) **Definition :** e is the number such that

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1.$$

$e \approx 2.7$ is an irrational number.

由(3.1-1)得 $(e^x)' = e^x$.

Example 1 : $(x^3 + 2x^2 - 3x + 1)' = 3x^2 + 4x - 3$.

Example 2 : $y = ae^x + \frac{b}{x}$
 $\Rightarrow y' = ae^x - \frac{b}{x^2}$.

Example 3 : Find the points on the curve

$$y = x^3 - x^2 - x + 1, \text{ where the tangent is horizontal.}$$

Solution : $y' = 3x^2 - 2x - 1 = (x-1)(3x+1) = 0$

$$\begin{array}{ccc} \Rightarrow x = 1 & \text{or} & x = -\frac{1}{3} \\ \Downarrow & & \Downarrow \\ y = 0 & & y = \frac{32}{27}. \end{array}$$

Answer : $(1, 0)$ and $(-\frac{1}{3}, \frac{32}{27})$.

Example 4 : Find $y = ax^2 + bx$ whose tangent at $(1, 1)$ has equation

$$y = 3x - 2.$$

Solution : Passing $(1, 1)$, $f'(1) = 3 \Rightarrow \begin{cases} a+b=1 \\ 2a+b=3 \end{cases} \Rightarrow a = 2, b = -1$.

Example5 : Find a cubic function $f(x) = ax^3 + bx^2 + cx + d$ whose graph has

horizontal tangents at $(-2, 6), (2, 0)$.

Solution: $f'(x) = 3ax^2 + 2bx + c$

$$\text{Passing } (-2, 6) \Rightarrow -8a + 4b - 2c + d = 6$$

$$\text{Passing } (2, 0) \Rightarrow 8a + 4b + 2c + d = 0$$

$$f'(2) = 0 \Rightarrow 12a + 4b + c = 0$$

$$f'(-2) = 0 \Rightarrow 12a - 4b + c = 0$$

$$\Rightarrow a = \frac{3}{16}, b = 0, c = -\frac{9}{4}, d = 3.$$

Example 6 : $\lim_{x \rightarrow 1} \frac{x^{2007} - 1}{x - 1} = ?$

Solution : $\lim_{x \rightarrow 1} \frac{x^{2007} - 1}{x - 1} = f'(1), \quad \text{where } f(x) = x^{2007}$
 $\Rightarrow f'(1) = 2007.$

Example 7 : Let $f(x) = \begin{cases} x^2 & \text{if } x \leq 2 \\ mx + b & \text{if } x > 2 \end{cases}$

Find the values of m and b that make f differentiable everywhere .

Solution : $f'(x) = \begin{cases} 2x & \text{if } x < 2 \\ m & \text{if } x > 2 \end{cases}$

$$\text{Continuous at } x = 2 \Rightarrow f(2^-) = f(2^+) \Rightarrow 4 = 2m + b.$$

$$\text{Differentiable at } x = 2 \Rightarrow f'(2^-) = f'(2^+) \Rightarrow 4 = m.$$

$$\Rightarrow b = -4.$$

Example 8 : Let $g(x) = \begin{cases} -1 - 2x & x < -1 \\ x^2 & -1 \leq x < 1 \\ x & x \geq 1 \end{cases}$

(i) Find the set of points at which g is continuous.

(ii) Find the set of points at which g is differentiable.

Solution : (i) $g(-1^-) = 1 = g(-1^+)$

$$g(1^-) = 1 = g(1^+)$$

$\Rightarrow$ The set of continuity of f is $\mathbb{R}$.

$$(ii) \quad g'(x) = \begin{cases} -2 & x < -1 \\ 2x & -1 < x < 1 \\ 1 & x > 1 \end{cases}$$

$$g'(-1^-) = -2 = g'(-1^+)$$

$$g'(1^-) = 2 \neq 1 = g'(1^+)$$

$\Rightarrow$ The set of differentiability of f is $\mathbb{R} - \{1\}$.